

LAB EXPERIMENTS

Tool: Python & R Programming

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Subject: Data Handling & Visualization

Code: DSA0604

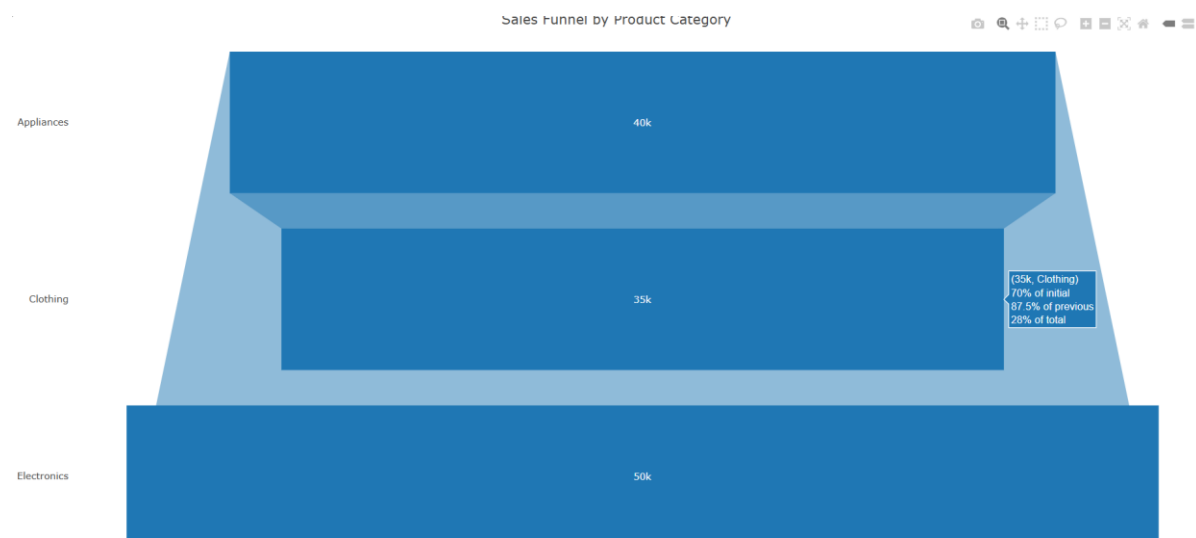
Set 11: Product Category Analysis

1. FUNNEL CHART

CODE:

```
library(plotly)
category <- c("Electronics","Appliances","Clothing")
sales <- c(50000,40000,35000)
plot_ly(
  type = "funnel",
  y = category,
  x = sales
) %>%
layout(title="Sales Funnel by Product Category")
```

OUTPUT:



2.TABLE

CODE:

```
category_data <- data.frame(  
  Category=c("Electronics","Clothing","Appliances"),  
  Sales=c(50000,35000,40000)  
)  
print(category_data)
```

OUTPUT:

```
> print(category_data)  
      Category Sales  
1 Electronics 50000  
2   Clothing 35000  
3 Appliances 40000  
> |
```

3.Dashboard (Pie + Funnel + Table)

CODE:

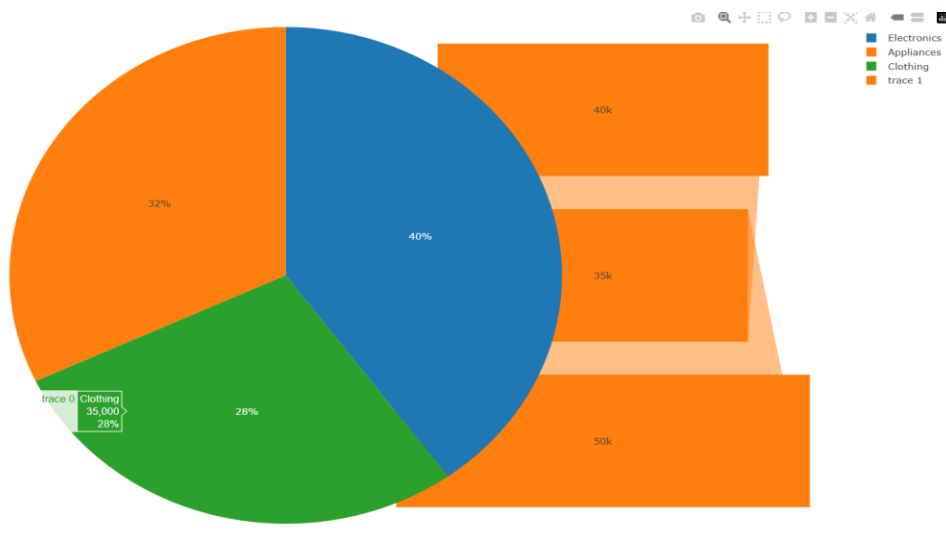
```
library(plotly)  
category <- data.frame(  
  Category = c("Electronics","Clothing","Appliances"),  
  Sales = c(50000,35000,40000)  
)  
pie_chart <- plot_ly(  
  category,  
  labels = ~Category,  
  values = ~Sales,  
  type = "pie"  
)  
funnel_chart <- plot_ly(  
  type = "funnel",  
  y = category$Category,  
  x = category$Sales  
)
```

```

subplot(
  pie_chart,
  funnel_chart,
  nrows = 1,
  margin = 0.08
)
print(category)

```

OUTPUT:



4. Pie Chart

CODE:

```

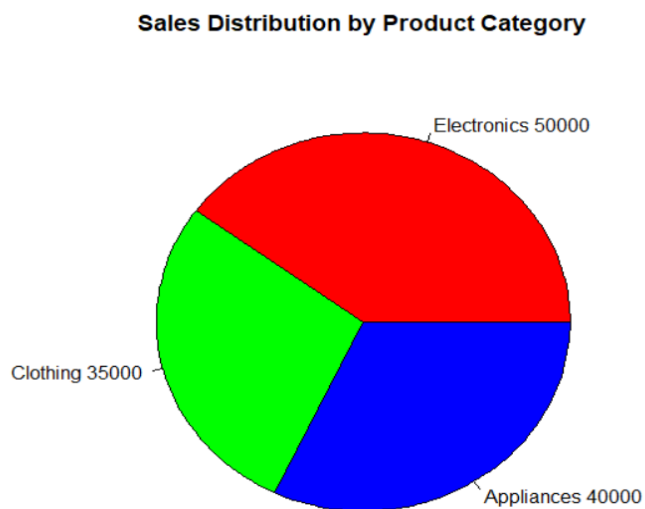
category <- data.frame(
  Category = c("Electronics","Clothing","Appliances"),
  Sales = c(50000,35000,40000)
)

pie(
  category$Sales,
  labels = paste(category$Category,
    category$Sales),
  col = rainbow(3),
  main = "Sales Distribution by Product Category"
)

```

)

OUTPUT:



SET 12: WEBSITE TRAFFIC

1. Line Chart (Daily Page Views)

CODE:

```
library(ggplot2)

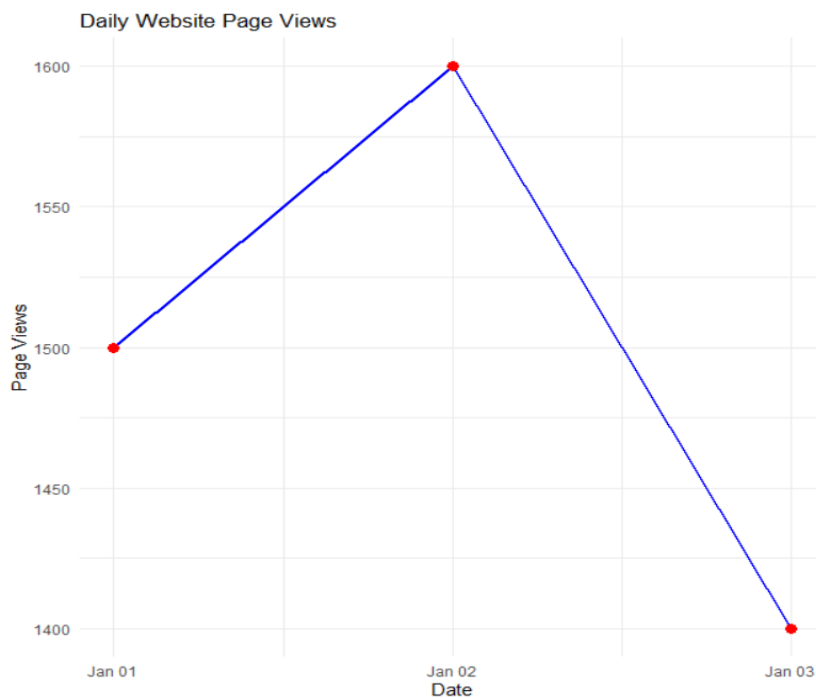
traffic <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  PageViews = c(1500, 1600, 1400)
)

ggplot(traffic,
  aes(x = Date,
    y = PageViews,
    group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Website Page Views",
    x = "Date",
    y = "Page Views"
```

```
) +
```

```
theme_minimal()
```

OUTPUT:



2: Bar Chart (Top Click-Through Rates)

CODE:

```
library(ggplot2)

traffic <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),
  CTR = c(2.3, 2.7, 2.0)
)

traffic <- traffic[order(-traffic$CTR), ]

ggplot(traffic,
  aes(x = reorder(Date, CTR),
    y = CTR,
    fill = Date)) +
  geom_col() +
  labs(
    title = "Top Click-Through Rates",
```

```

x = "Date",
y = "CTR (%)"
) +
theme_minimal()

```

OUTPUT:



3. Table (Website Traffic Data)

CODE:

```

traffic <- data.frame(
  Date = c("2023-01-01","2023-01-02","2023-01-03"),
  PageViews = c(1500,1600,1400),
  CTR = c("2.3%","2.7%","2.0%")
)
print(traffic)

```

OUTPUT:

```

> print(traffic)
  Date PageViews  CTR
1 2023-01-01    1500 2.3%
2 2023-01-02    1600 2.7%
3 2023-01-03    1400 2.0%
>

```

4. Dashboard (Line Chart + Bar Chart + Table)

CODE:

```
library(ggplot2)
library(patchwork)

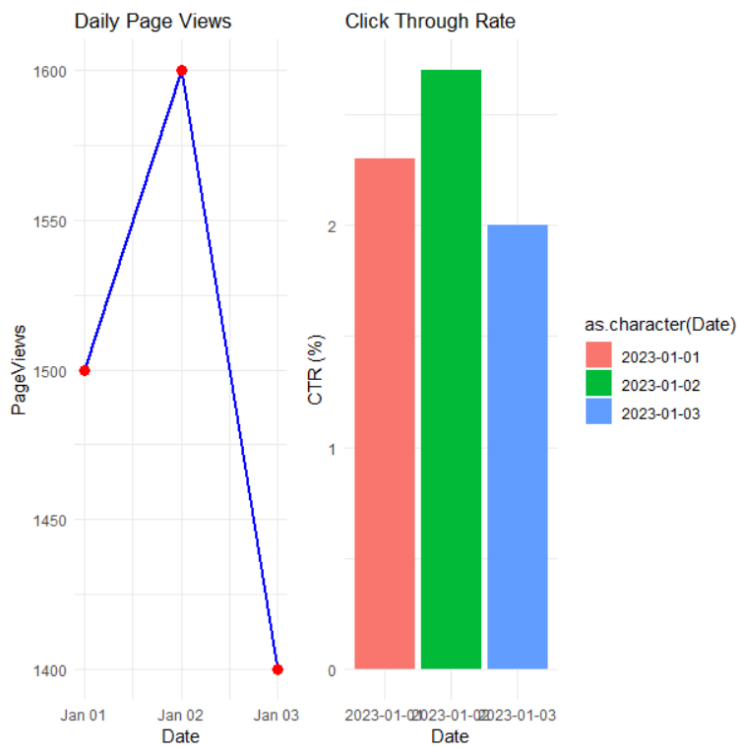
traffic <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  PageViews = c(1500, 1600, 1400),
  CTR = c(2.3, 2.7, 2.0)
)

p1 <- ggplot(traffic,
  aes(Date, PageViews, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(title = "Daily Page Views") +
  theme_minimal()

p2 <- ggplot(traffic,
  aes(as.character(Date), CTR, fill = as.character(Date))) +
  geom_col() +
  labs(title = "Click Through Rate",
    x = "Date",
    y = "CTR (%)") +
  theme_minimal()

p1 + p2
print(traffic)
```

OUTPUT:



SET 13: GEOGRAPHIC DATA

1. Map Chart

CODE:

```
library(ggplot2)

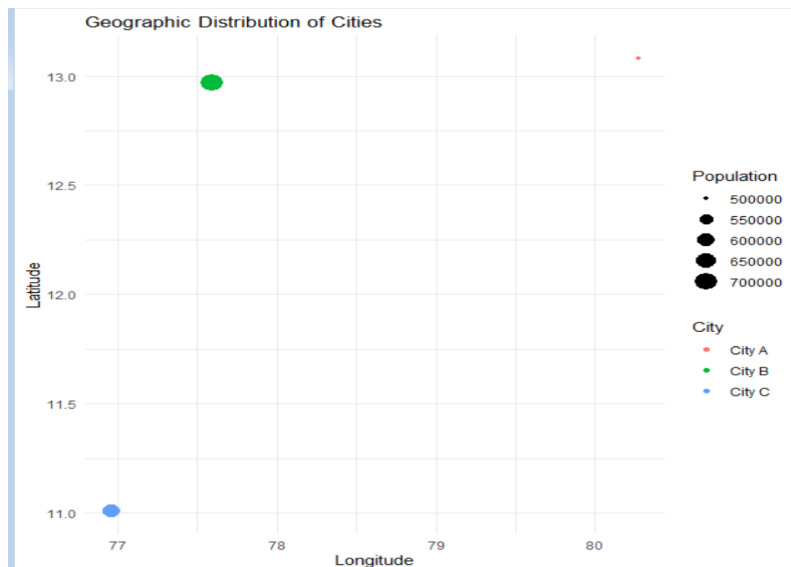
geo <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200),
  Latitude = c(13.08, 12.97, 11.01),
  Longitude = c(80.27, 77.59, 76.96)
)

ggplot(geo,
  aes(x = Longitude,
    y = Latitude,
    size = Population,
    color = City)) +
```



```
geom_point() +
labs(title="Geographic Distribution of Cities",
      x="Longitude",
      y="Latitude") +
theme_minimal()
```

OUTPUT:



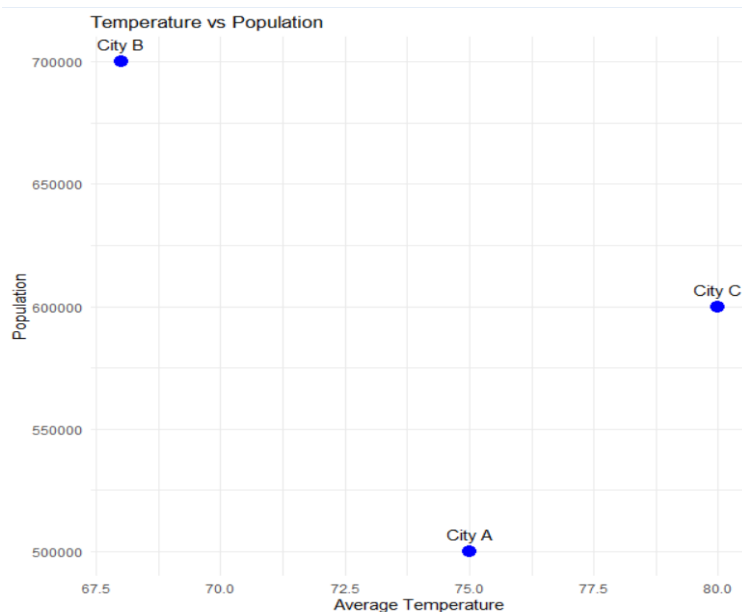
2. Scatter Plot (Temperature vs Population)

CODE:

```
library(ggplot2)
geo <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Temperature = c(75, 68, 80)
)
ggplot(geo,
  aes(x=Temperature,
      y=Population,
      label=City))+
  geom_point(color="blue",size=4)+
  geom_text(vjust=-1)+
```

```
labs(title="Temperature vs Population",
     x="Average Temperature",
     y="Population")+
theme_minimal()
```

OUTPUT:



3.Table

CODE:

```
geo <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)
print(geo)
```

OUTPUT:

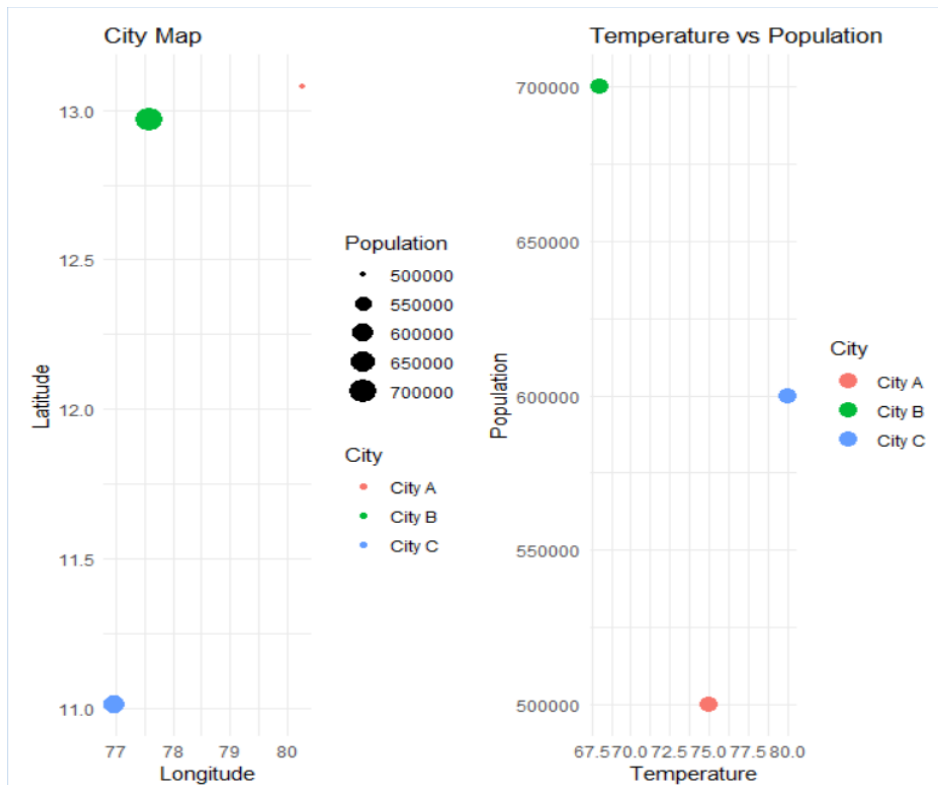
```
> print(geo)
  City Population Temperature Elevation
1 City A      5e+05         75       1000
2 City B      7e+05         68         800
3 City C      6e+05         80       1200
> |
```

4.Dashboard (Map + Scatter + Table)

CODE:

```
library(ggplot2)
library(patchwork)
geo <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200),
  Latitude = c(13.08, 12.97, 11.01),
  Longitude = c(80.27, 77.59, 76.96)
)
p1 <- ggplot(geo,
  aes(Longitude, Latitude,
    size=Population,
    color=City))+
  geom_point()+
  ggtitle("City Map")+
  theme_minimal()
p2 <- ggplot(geo,
  aes(Temperature,
    Population,
    color=City))+
  geom_point(size=4)+
  ggtitle("Temperature vs Population")+
  theme_minimal()
p1 + p2
print(geo)
```

OUTPUT:



SET 14: SURVEY RESULTS

1. Stacked Bar Chart

CODE:

```
library(ggplot2)

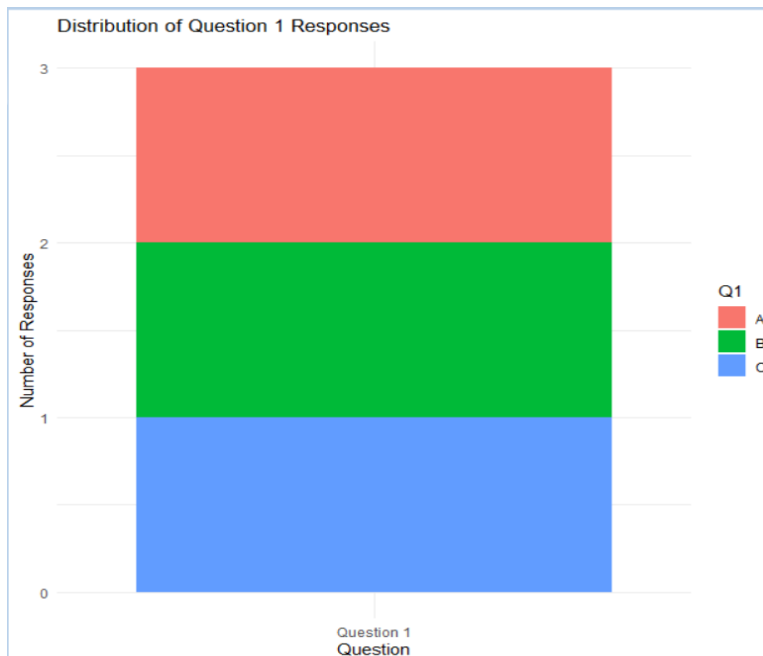
survey <- data.frame(
  Respondent = c(1,2,3),
  Q1 = c("A","B","C")
)

ggplot(survey,
  aes(x = "Question 1",
    fill = Q1)) +
  geom_bar() +
  labs(
    title = "Distribution of Question 1 Responses",
    x = "Question",
    y = "Number of Responses"
```

```
) +
```

```
theme_minimal()
```

OUTPUT:



2. Radar Chart

CODE:

```
library(fmsb)

data <- data.frame(
  A = c(5,2,3),
  B = c(5,1,2),
  C = c(5,3,1),
  D = c(5,0,1)
)

rownames(data) <- c("Max", "Min", "Responses")

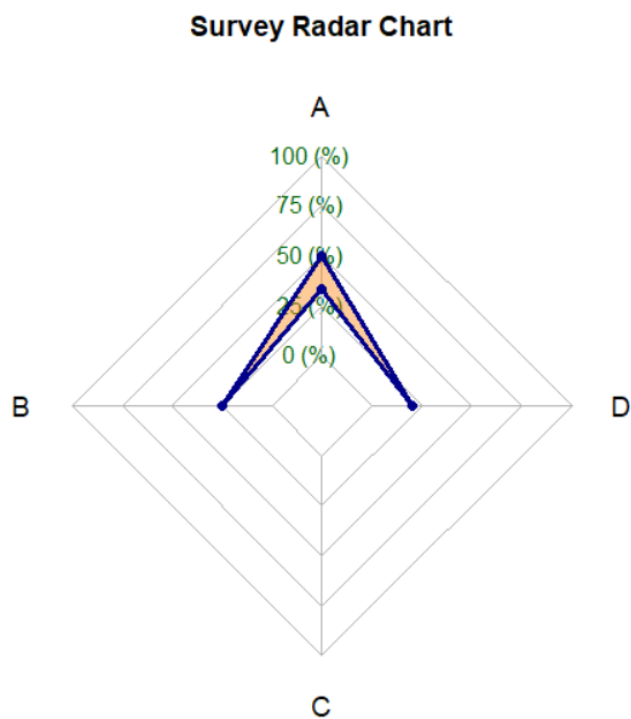
radarchart(
  data,
  axistype = 1,
  pcol = "darkblue",
  pfc = rgb(1,0.5,0,0.4), # Light Orange Fill
  plwd = 3,
```

```

plty = 1,
cglcol = "grey70",
cglty = 1,
cglwd = 1,
axislabcol = "darkgreen",
vlcex = 1.2,
title = "Survey Radar Chart"
)

```

OUTPUT:



3.Table

CODE:

```

survey <- data.frame(
  Respondent = c(1,2,3),
  Q1 = c("A","B","C"),
  Q2 = c("B","A","A"),
  Q3 = c("C","D","B")
)

```

```
print(survey)
```

OUTPUT:

```
  Respondent Q1 Q2 Q3  
1           1  A  B  C  
2           2  B  A  D  
3           3  C  A  B  
> |
```

4. Dashboard (Stacked Bar + Radar + Table)

CODE:

```
library(ggplot2)
```

```
library(patchwork)
```

```
library(fmsb)
```

```
survey <- data.frame(  
  Respondent = c(1,2,3),
```

```
  Q1 = c("A","B","C"),
```

```
  Q2 = c("B","A","A"),
```

```
  Q3 = c("C","D","B")
```

```
)
```

```
p1 <- ggplot(survey,
```

```
  aes(x="Question 1",
```

```
  fill=Q1)) +
```

```
  geom_bar() +
```

```
  ggtitle("Question 1 Responses") +
```

```
  theme_minimal()
```

```
print(p1)
```

```
data <- data.frame(  
  A=c(5,2,3),
```

```
  B=c(5,1,2),
```

```
  C=c(5,3,1),
```

```
  D=c(5,0,1)
```

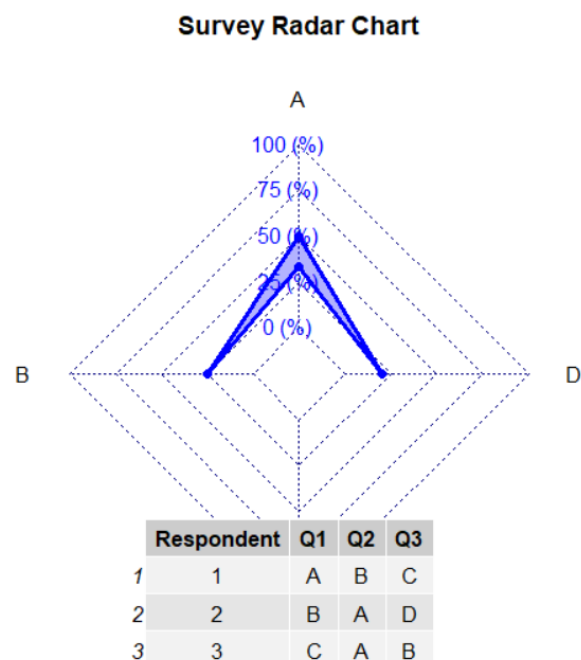
```
)
```

```

rownames(data)=c("Max","Min","Responses")
radarchart(
data,
axistype=1,
pcol="red",
pfc="pink",
plwd=2
)
print(survey)

```

OUTPUT:



SET 15: PRODUCT SALES ANALYSIS

1.Grouped Bar Chart

CODE:

```

library(ggplot2)
sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),

```

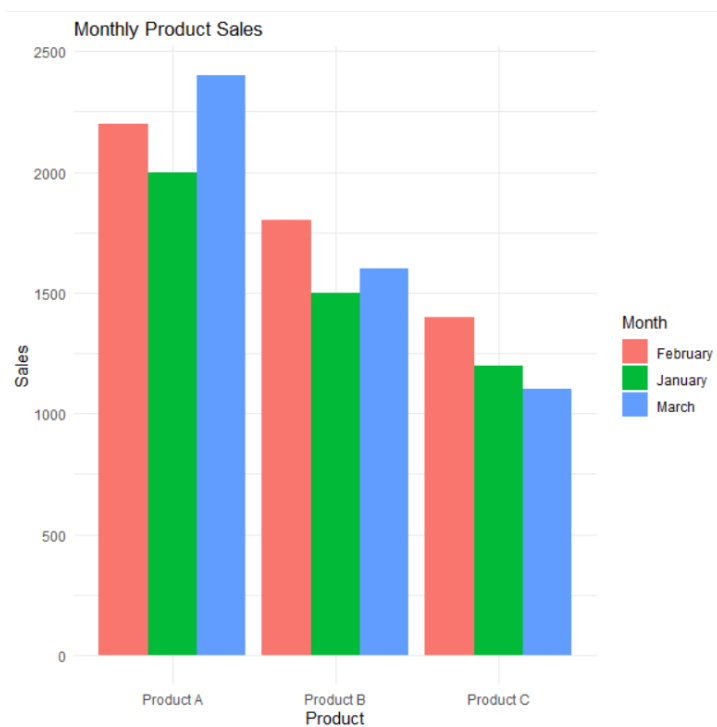


```

March = c(2400,1600,1100)
)
sales_long <- data.frame(
  Product = rep(sales$Product,3),
  Month = rep(c("January","February","March"), each=3),
  Sales = c(sales$January,sales$February,sales$March)
)
ggplot(sales_long,
  aes(x=Product,
    y=Sales,
    fill=Month))+
  geom_bar(stat="identity",
    position="dodge")+
  labs(title="Monthly Product Sales",
    x="Product",
    y="Sales")+
  theme_minimal()

```

OUTPUT:



2.Stacked Area Chart

CODE:

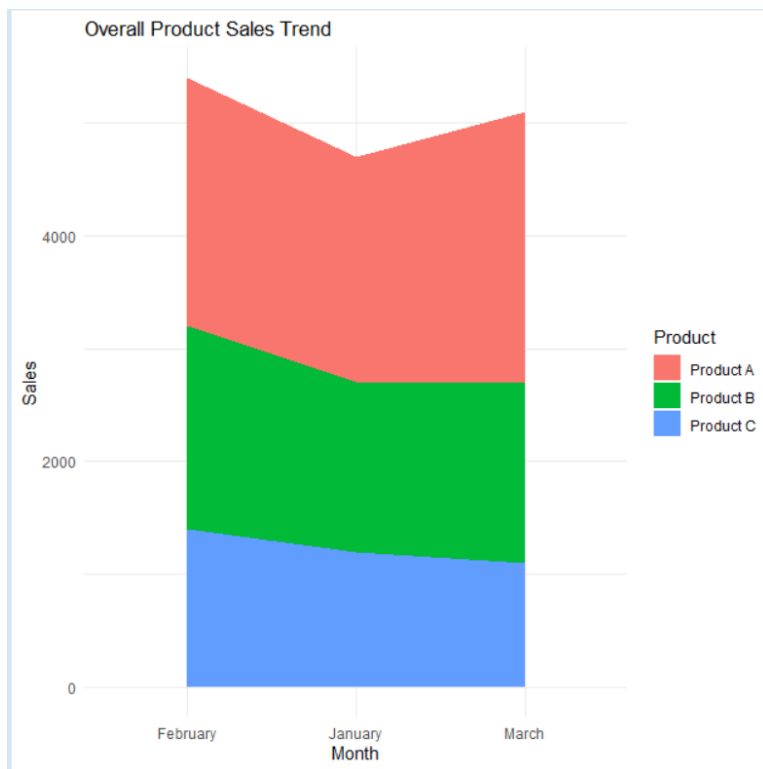
```
library(ggplot2)

sales <- data.frame(
  Month=c("January", "February", "March"),
  ProductA=c(2000,2200,2400),
  ProductB=c(1500,1800,1600),
  ProductC=c(1200,1400,1100)
)

area_data <- data.frame(
  Month=rep(sales$Month,3),
  Product=rep(c("Product A", "Product B", "Product C"),each=3),
  Sales=c(sales$ProductA,
          sales$ProductB,
          sales$ProductC)
)

ggplot(area_data,
  aes(x=Month,
      y=Sales,
      fill=Product,
      group=Product))+
  geom_area()+
  labs(title="Overall Product Sales Trend",
       x="Month",
       y="Sales")+
  theme_minimal()
```

OUTPUT:



3.Table

CODE:

```
sales <- data.frame(
  Product=c("Product A","Product B","Product C"),
  January=c(2000,1500,1200),
  February=c(2200,1800,1400),
  March=c(2400,1600,1100)
)
print(sales)
```

OUTPUT:

```

# A tibble: 3 x 4
#   Product January February March
#   <fct>   <dbl>   <dbl> <dbl>
1 Product A    2000    2200  2400
2 Product B    1500    1800  1600
3 Product C    1200    1400  1100
> |
```

4. Dashboard (Grouped Bar Chart + Stacked Area Chart + Table)

CODE:

```
library(ggplot2)
library(patchwork)
```

```

sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

sales_long <- data.frame(
  Product = rep(sales$Product,3),
  Month = rep(c("January","February","March"), each=3),
  Sales = c(sales$January,sales$February,sales$March)
)

p1 <- ggplot(sales_long,
  aes(Product,
    Sales,
    fill=Month))+
  geom_bar(stat="identity",
    position="dodge")+
  ggtitle("Grouped Bar Chart")+
  theme_minimal()

area_data <- data.frame(
  Month=rep(c("January","February","March"),3),
  Product=rep(c("Product A","Product B","Product C"),each=3),
  Sales=c(2000,2200,2400,
    1500,1800,1600,
    1200,1400,1100)
)

p2 <- ggplot(area_data,
  aes(Month,
    Sales,

```

```

    fill=Product,
    group=Product))+
  geom_area()+
  ggtitle("Stacked Area Chart")+
  theme_minimal()

p1 + p2
print(sales)

```

OUTPUT:

